Predicting $PM_{2.5}$ Concentrations Across North America

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I. ABSTRACT

This paper explores the monthly forecasting of $PM_{2.5}$ concentrations across 1 km grid-cells within North America, measured in micrograms per cubic meter ($\mu g/m^3$), by using various machine learning methods. Known as a very dangerous form of pollution, particulate matter, especially small particles, can cause serious health problems[7]. Its concentrations are also extremely variable, and very little is known/publicly accessible about what the future may look like. Previous studies, however, have found that two of the main determinants for $PM_{2.5}$ concentrations include location and time[1]. Therefore, throughout this paper the spatiotemporal relationships of $PM_{2.5}$ will be covered as to how they affect detailed prediction of future levels.

The results will enable not only those in North America to gain an understanding of what the future may look like but provide a solid foundation for large-scale forecasting. The first dataset, retrieved from our parent paper titled, *First close insight into global daily gapless 1 km $PM_{2.5}$ pollution, variability, and health impact[1]*, provides a comprehensive collection of $PM_{2.5}$ measurements worldwide. Specifically, it included $PM_{2.5}$ values for every square kilometer over every landmass (distinguished by latitude and longitude coordinates), through the years 2017-2022. These measurements were collected through the combination of ground measurements, satellite aerosol optical depth (AOD) measurements, as well as chemical transport model predictions[1]. The next dataset titled ERA5 was a worldwide reanalysis of meteorological factors. Using these global datasets, a boundary was first set using longitude and latitude coordinates that captured North America, making sure to remove the large counts of missing data across bodies of water. This distinction between North America, as opposed to covering the entirety of the world, provides a more relevant and focused output that could in turn provide the most benefit.

Realizing the risk $PM_{2.5}$ imposes on humanity, any accurate information that can be learned about the future would provide immeasurable benefit in terms of public health decisions, understanding climate change, urban planning, and many more. Next, various ensemble/advanced ML methods were trained to bridge the gap from past historical prediction towards a more significant forecasting model. Some of these models included a Naïve Baseline, Linear Regression, CatBoost Regression[3], XGBoost[3], and LightGBM[12]. Model performance was then evaluated under metrics like RMSE, MAE, R^2, and Bias. As will be discussed in detail, the gradient boosting methods were consistently able to capture variations in $PM_{2.5}$ across both space and time, whereas the LR and Naïve models struggled.

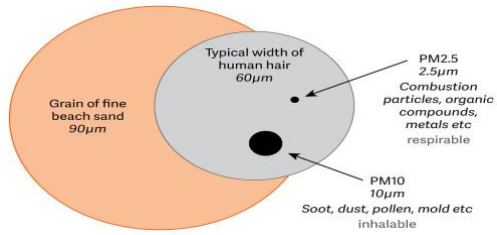

Figure 1 - Comparison between size of $PM_{2.5}$ Particle and Strand of Hair[4]

II. INTRODUCTION

Air quality is an important topic worldwide, and air pollution is a leading global risk factor, causing 6.7

million deaths each year[1]. Out of all the dangerous air pollution particles, $PM_{2.5}$ is one of the most dangerous[7]. This is due to its extremely small size: one $PM_{2.5}$ particle is more than 20 times thinner than a strand of hair, or less than 2.5 micrometers in diameter[4]. More specifically, its size makes it dangerous because it can make its way into the body much easier, and in higher quantities than many other air pollutants. This can be very dangerous for both cardiovascular and respiratory health[7]. It is important to be able to understand and predict these levels so that the problem of $PM_{2.5}$ pollution can be prevented. By having an accurate predictive model, preventative policies like limitations on factory emissions and limits on greenhouse gases can be put in place[8]. Additionally, accurate predictions of $PM_{2.5}$ pollution can be used to send out personalized alerts to those at risk. Our research aimed to answer which models are the most effective for both historical $PM_{2.5}$ predictions and forecasting future concentrations. The addition of meteorological data was also explored to determine if it could benefit performance and if so, to what extent.

Currently, there is a large data gap. Satellites provide a lot of pollution data, but cloud cover, as well as snow and ice, create large data gaps across the globe[1]. Additionally, ground monitoring stations are often spread too far apart to be able to capture localized pollution hotspots[1]. Although many models in the past have been able to predict well against historical data, there is a need for real-time prediction and future forecasting.

Spatiotemporal modelling has already been done through Geographically and Temporally Weighted Regression and Random Forest Regression as seen in some of the other referenced papers[2]. These were able to model the complex relationships between geography and air quality well. Our parent paper was also able to aggregate satellite data, ground-monitoring data, and other auxiliary data to create the first global, daily, gapless $PM_{2.5}$ dataset using a 4D-STET model[1]. This 4-dimensional model can be interpreted as one that takes latitude, longitude, time, and input $PM_{2.5}$ measurements to map out expected and detailed levels of concentrations of $PM_{2.5}$ throughout the world. Additionally, there was a study done in India where they predicted concentrations of different pollutants in four cities using more advanced machine learning methods like CatBoost and XGBoost to handle imbalanced and non-linear data[3].

These past studies provided insight into how $PM_{2.5}$ behaves and its potential impact; however, they were only based on historical predictions. While it is important to understand how $PM_{2.5}$ might have behaved in the past, given it allows to make inferences about the future, the transition to forecasting $PM_{2.5}$ can mean the difference between simply understanding why lives were lost in the past to directly saving lives in the future. This paper develops a refined machine learning approach using Ridge Regression, XGBoost, CatBoost and LightGBM for future forecasting of $PM_{2.5}$ concentrations across North America. The models use a down sampled version of the 1 km resolution gapless dataset, while still ensuring that no information is lost due to cloud cover, snow, and ice. The models implement spatial neighbor lags and temporal rolling averages to capture the spatiotemporal movement of pollution across the continent.

Figure 2 below exhibits the chosen model pipeline for Forecasting $PM_{2.5}$ concentrations across North America. After having completed data preprocessing and finalizing our methodology, the training of the models began on monthly $PM_{2.5}$ measurements from 2017 through 2020. Then validating predictions in the year 2021 and finally testing them on the year 2022. While in an ideal world the data would have been slightly more recent, the reproducible foundations of this research open the door for future forecasting development. While the concentrations for the year 2023 were confidently predicted, generalizing predictions further in the future would have been less realistic given the large temporal gap that would need to be covered. The volatility of $PM_{2.5}$ over time is very important to note, as many extreme, unpredictable events like wildfires can cause unforeseen spikes[1]. Further, while the use of an entire year's worth of validation and testing data limits the overall data available for training, it ensures the model's outputs were generalizable. In short, this pipeline and project have massive potential for extended application, especially through the addition of $PM_{2.5}$ measurement inputs.

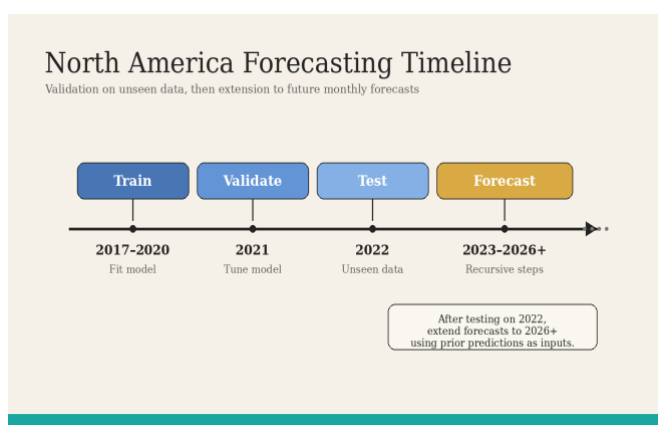

Figure 2 – Our forecasting flowchart for $PM_{2.5}$ concentrations

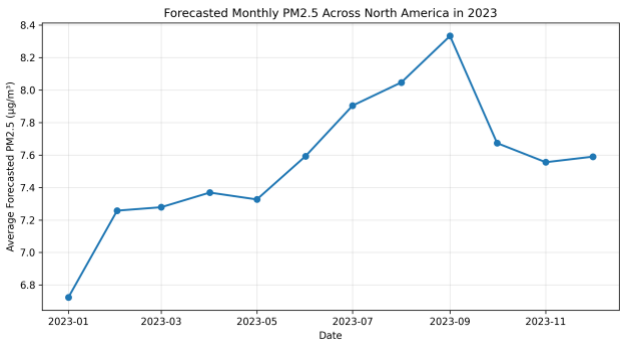

Figure 3 - Monthly Forecasted $PM_{2.5}$ Concentrations in North America utilizing CatBoost

As shown in Figure 3 above, there is an example outputted CatBoost Monthly Forecast of $PM_{2.5}$ concentrations averaged throughout North America. As shown in the graph, the trends are very inconsistent with expected spikes of $PM_{2.5}$ later in the year. Further, these predictions will be explored as to how they generalize on historical data, then reposition our predictions towards 2023. Given our data only extends until 2022, it is important to note that the further the models generalize or predict into the future, the higher the risk that the predictions are inaccurate. Especially with a topic as inconsistent as $PM_{2.5}$, the confidence level in produced outputs decreases as the models are forced to predict further into the future.

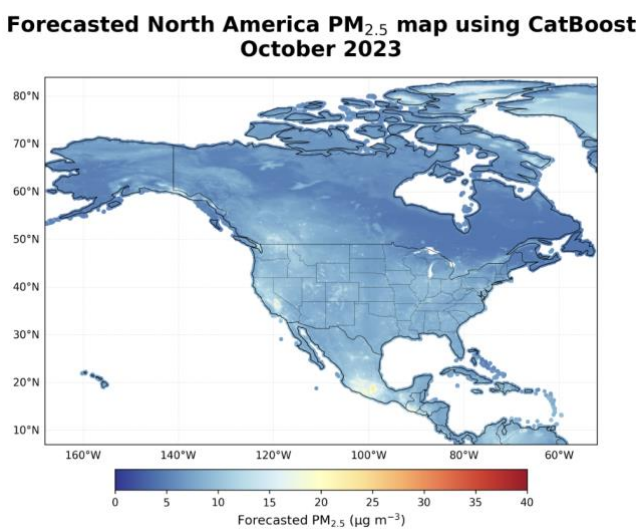

Figure 4 – Forecasted NA $PM_{2.5}$ map for October, 2023

Figure 4 also gives an idea of what our forecasting model(s) aimed to produce. Specifically showing some inconsistencies in $PM_{2.5}$ levels, with lighter more populated regions like central California and Mexico, along with places on the East Coast having higher $PM_{2.5}$ levels.

III. RELATED WORK

J. Wei et al.[1] created the first global daily gapless $PM_{2.5}$ dataset through a developed 4D-STET model: utilizing Satellite AOD measurements, ground monitoring, and chemical transport models to determine said $PM_{2.5}$ levels. They then broke down their findings by geographical location and explored how different regions contributed to various levels. This meant comparing rural vs. urban areas, as well as expanding this idea in discussing how developing countries compared to those already developed. They also examined COVID-19 and its direct effects on air pollution, including how those differences determined the mortality burden. Y. Shen et al.[2] investigated the modeling of Europe-wide air pollution over the course of 2000-2019, through geographically weighted regression. They specifically predicted $PM_{2.5}$, PM_{10}, NO_2, and O_3, making sure to compare how GWR compared to a GWR model with the added temporal component (GTWR). What they found was that spatial location plays a key role in air pollution concentrations, and that it is important that models can capture the non-linear relationships within pollution patterns. N. S. Gupta et al.[3] explored the prediction of air quality index in 4 major Indian cities, by looking at predicted levels of 11 different major air pollutants. They then categorized their results by levels such as good, satisfactory, moderate, poor, very poor, and severe, ensuring that their findings would remain interpretable by the most amount of people. They conducted their study through the implementation of Random Forest, Support Vector Regression, and CatBoost Regression, but importantly included data balancing through the employment of synthetic minority oversampling technique or SMOTE. They were able to find that not only does SMOTE improve accuracies but found CatBoost to be an important predictor of air pollution.

IV. NOVELTY

Prior studies in the field of Air Quality Monitoring and Prediction have consistently yielded high accuracy models that focus heavily on historical prediction. While it is important to learn about the past, it can be argued in the case of air quality, the future is just as, if not more important. Our paper aimed to break through this boundary of unknowns and decipher how air pollution, specifically $PM_{2.5}$ concentrations, change over time and space. Through the help of our related works above, and supplemental research, our study explored how previously acclaimed gradient boosting modeling methods, like XGBoost, CatBoost, and LightGBM, were able to make this transition into future forecasting. Not only did our methodology revolve around a more optimistic goal for future understanding but did so at both an extremely large-scale and resolution. Our final output was forecasted $PM_{2.5}$

concentrations, under a variety of models, for many 1 km square locations within North America.

Our models are not just learning off past pollution levels, they are painting a picture of how air pollution behaves in the environment and how that reflects $PM_{2.5}$ concentrations. This was achieved through the integration of meteorological variables obtained in the ERA5[5] dataset, a global weather dataset produced by the European Centre for Medium-Range Weather Forecasts (ECMWF)[5]. Some of these factors included temperature, precipitation, wind, surface pressure, overall cloud cover, and more. As a result, our research builds upon historical work done in terms of Air Quality in North America and sets the foundation for a future in which humanity understands the risks associated with the air we breathe.

V. METHODS

a. OVERVIEW

The methodology of our research was founded in first identifying the overarching limitations from our parent paper.[1] The main area for improvement was determined to be the need for a forecasting model. While it is important to understand past $PM_{2.5}$ patterns and values through historical prediction, getting an idea about the future can provide even more benefit. It is worth noting that the decision to focus on North America, rather than repeat our parent paper's focus on the entire world, was because the results would be more relevant and effective if they were centralized. While modeling the entire world sounds helpful, it can produce a lot of noise as you are trying to predict an extremely variable value over drastically different climates, geographical locations, population densities, etc. Focusing on a specific continent allowed our research to maintain scale while promoting impact. Further, monthly forecasts were pursued over shorter time scales, as they enabled a cheaper and quicker assessment of $PM_{2.5}$ patterns, while still retaining a high level of seasonal and temporal variation.

b. DATA PREPROCESSING

First, the global monthly $PM_{2.5}$ dataset, which included $PM_{2.5}$ values in $\mu g/m^3$ for the entire globe, was subsetted to a chosen latitude and longitude box (coordinates listed in DATA AVAILABILITY). The rows recording $PM_{2.5}$ values over bodies of water (where $PM_{2.5} = 0$) were then removed as our dataset only included positive levels over Earth's landmasses. The resulting 23,160,888 rows were then stored as a CSV, each containing a Latitude, Longitude, $PM_{2.5}$ Concentration, and a Year-Month timestamp. Due to very high computational costs, only every 10th latitude and longitude coordinates were selected, meaning our entire dataset would have been roughly 2.3 billion rows in length without this downsample. Additionally, our global weather ERA5 dataset[5] was incorporated by again subsetting to cover North America and our chosen timeline (2017-2022). With the weather data being organized by latitude and longitude and assigned relevant information to their closest spatial $PM_{2.5}$ measurement. This enabled the models to seamlessly train on multiple data sources.

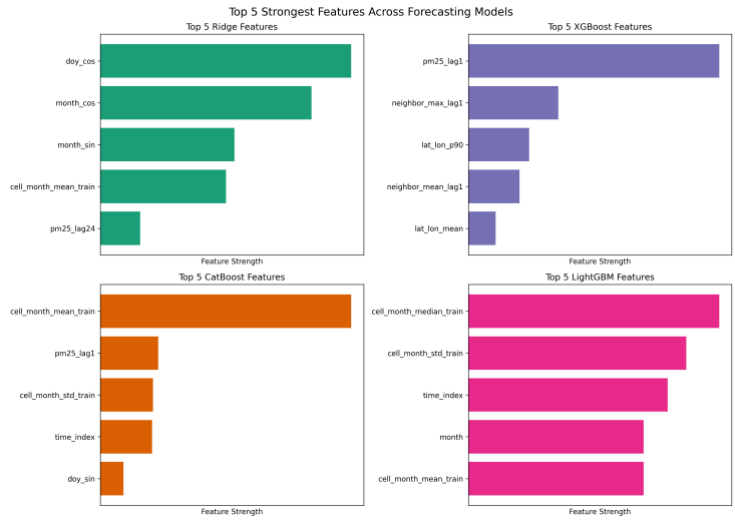

Figure 5 – All Model Feature Importance Chart

c. FEATURE ENGINEERING

To extract the most information from our dataset a large amount of feature combinations and interactions were explored. Our research heavily relied on these created relationships to predict future concentrations of $PM_{2.5}$. Some of the most important features through the models, as shown in Figure 5 above, were time indices, lags (short and longer-term), historical cell information (mean, standard deviation, median), and spatial or neighboring cell information. When it came to the addition of ERA5 features, they were purely included for the improvement of historical predictions. Given that ERA5 is a reanalysis[5], its level of detail and information was not practical to incorporate in the forecasting implementation.

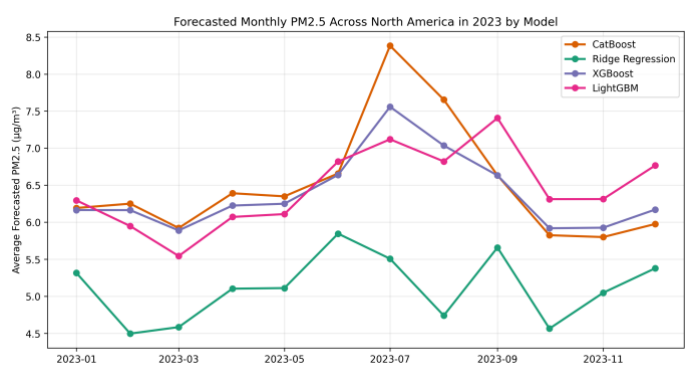

Figure 6 – Forecasted Monthly Averages for the 4 Developed Models

Figure 6 contains the outputted forecasted averages of $PM_{2.5}$, per month, in $\mu g/m^3$ for the four models. It exemplifies a product of our research and the overarching goal to form an understanding about how $PM_{2.5}$ levels can be predicted and

ways in which that will impact humanity.

d. MODELS

Throughout our research on $PM_{2.5}$, gradient boosting methods were found to be particularly successful in parsing through high-dimensional data and determining accurate pollution concentrations. Our chosen baseline was a Naïve model that would essentially predict $PM_{2.5}$ values for each latitude and longitude combination based off its previous value. As expected, the Naïve model struggled with understanding the complex trends in $PM_{2.5}$ values. Building on these predictions, models like Ridge Regression, XGBoost, CatBoost, and LightGBM were explored. It is worth mentioning that the target predictions from the tree-based ensemble models were transformed through the Log1p scale, which essentially acts as a safer logarithmic function. It calculates Log(1+x) for every prediction, ensuring that near-zero or zero values don't become undefined as Log(0) outputs.

Ridge Regression

Ridge Regression was implemented after the Naïve model in the hopes of serving as a more advanced baseline. It primarily functioned as an advanced linear regression model that could understand and interpret certain spatiotemporal features. Ridge Regularization works by weighting and penalizing certain features to ensure that they don't heavily influence predictions. This regularization helped to limit overfitting and at the same time demonstrate the larger amount of $PM_{2.5}$ variability that can be explained by simple features alone. This specific model was chosen because the more advanced the baseline outputs were, the more meaningful specific model changes would appear in terms of predictive improvements. Further, while Ridge Regression improved upon the Naïve model's predictions, it had trouble interpreting more complex feature interactions. Importantly, the model struggled due to its use of singular coefficients and assumptions that localized spikes in $PM_{2.5}$ would be consistent in the future.

XGBoost

Next, a widely used air quality model in XGBoost was implemented. In terms of our research XGBoost functioned as a nonlinear model that would build a set of trees with $PM_{2.5}$ predictions and check whether these predictions were right. Assuming error existed, it would then continuously create more trees and compare results to limit inaccuracies. This model was chosen because of its ability to handle how coefficients of features like seasons, temperatures, locations, may change over time. Not only can XGBoost understand these interactions, but it can also do so at extremely high speeds. Throughout our validation and testing phases which are covered in the RESULTS AND CONCLUSIONS section, XGBoost consistently excelled among our models for predicting $PM_{2.5}$ concentrations across North America.

CatBoost

As covered in our 3rd research paper, CatBoost, was found to be a gradient boosting method found to rival XGBoost in terms of speed and accuracy. The study examined AQI predictions in 4 Indian cities and found XGBoost to perform better for 2 of the cities, while CatBoost was the better choice for the remaining cities[3]. While the methods work very similarly, an important difference is how they handle features. In our research CatBoost viewed information like month, or our created latitude/longitude bins feature, as categorical, whereas XGBoost handled them as numerical inputs. This model was essentially created to address whether this ideology shift had any benefit in terms of predictive capability.

LightGBM

Our 5th and final model was LightGBM, which functions similarly to the previous two tree-based gradient boosting methods. A key benefit of LightGBM, however, is its training speed and ability to handle larger datasets. These factors make LightGBM particularly favorable due to our $PM_{2.5}$[1] and ERA5[5] datasets being extremely large, and model training taking extended amounts of time. LightGBM was tasked with answering the question of whether a more heavy-duty ensemble model was needed to improve both prediction and the forecasting of $PM_{2.5}$ concentrations.

e. EVALUATION

As mentioned in DATA PREPROCESSING, our study aimed to have our model learn over the years 2017-2022, with the training phase being on 2017-2020 data, then validating those predictions over 2021 data, and finally testing in 2022. It is important to recognize that this yearly split can carry confounding variables when it comes to $PM_{2.5}$ levels, given that these levels can change on a yearly basis. For example, an article published in the National Library of Medicine conducted a study on $PM_{2.5}$ and claimed that its concentrations primarily decreased within the contiguous United States during COVID-19. The article writes, "most of the urban environments — Washington DC, New York, Boston, Chicago, Los Angeles, Houston, Dallas, Philadelphia, Detroit, and Seattle — show a decrease in $PM_{2.5}$ levels by 21.1%, 20.7%, 18.5%, 8.05%, 3.29%, 3.63%, 6.71%, 4.82%, 13.5%, and 7.73%, respectively, between March–May of 2020 and 2019"[6] (Masoud G. et al., 2022). This idea reveals that locations can have fluctuations in $PM_{2.5}$ concentrations from external unpredictable influences. Therefore, it is important to realize that when it comes to our research, accurately and consistently predicting pollution levels is very challenging given all the unknowns surrounding particulate matter.

After models were thoroughly trained, validated, and tested on the yearly splits, they were evaluated under four key metrics to compare their predictive performances. These metrics included R^2, RMSE, MAE, and Bias, as well as comparing model predictions with and without the added ERA5 features to determine if it made a meaningful change. Building on those evaluation metrics, RMSE and MAE are among some of the most important in terms in ensuring the models are generalizable and that their methodology can be reused to forecast $PM_{2.5}$.

Test-Set Forecasting Performance Across All Models

Model	Scenario	RMSE	MAE	R2
Naive	lag1 baseline	2.385	1.297	0.350
Ridge Regression	Without ERA5	2.307	1.550	0.392
XGBoost	Without ERA5	1.844	1.226	0.611
CatBoost	Without ERA5	1.793	1.159	0.633
LightGBM	Without ERA5	1.867	1.221	0.601

Figure 7 – Evaluation Metrics for All Models without ERA features

VI. RESULTS AND CONCLUSIONS

a. RESULTS

Above you can see the test evaluation metrics for the five models over the year of 2022, without the added ERA5 dataset[5]. Looking at the table, it can clearly be seen that CatBoost performed the best as it held the lowest RMSE, lowest MAE, and highest R^2 value across the models. The R^2 value of 0.633 or 63.3% indicates that it was able to explain a large portion of the variability of $PM_{2.5}$ concentrations across North America for 2022. Further, the RMSE and MAE values of 1.793 & 1.159 respectively, show that the model was also the most generalizable. In other words, CatBoost was the best choice for forecasting as it featured fewer total errors, and fewer large prediction errors within the year 2022. As expected, the Naïve model performed the worst on average of the models, with Ridge regression performing the second worst. This reaffirms the idea that $PM_{2.5}$ is very complicated to understand and predict over both time and space. Only when complex feature engineering was implemented could impactful estimates for $PM_{2.5}$ levels in microgram per m^3 be achieved. Additionally, on average our R^2 values increased by approximately 43.1% across the gradient boosting models, when compared to the original Naïve baseline R^2 of 0.350. In summary, the CatBoost model was the best overall framework for predicting pollution without ERA5 features being used.

Test-Set Forecasting Performance Across All Models With ERA5

Model	Scenario	RMSE	MAE	R2
Naive	lag1 baseline	2.385	1.297	0.350
Ridge Regression	With ERA5	2.221	1.507	0.436
XGBoost	With ERA5	1.878	1.216	0.597
CatBoost	With ERA5	1.876	1.230	0.598
LightGBM	With ERA5	1.846	1.196	0.610

Figure 8 – Evaluation Metrics for All Models with ERA features

In the hopes of further improvement, a dataset including the reanalysis of weather measurements, was fed into the models through large-scale feature engineering. The original baseline models of Ridge Regression, CatBoost, XGBoost, and LightGBM involved 18, 40, 51, and 51 total features respectively without ERA. Once the weather data was inputted that number increased to 36, 58, 69, and 69 features. It is worth noting that of the gradient boosting approaches (before ERA), CatBoost was able to achieve the most with the least number of features, further exhibiting its better performance of the tested models. Moreover, the addition of features increased the complexity of the models dramatically, and looking at the results, that meant different things for different methods. For example, out of the 4 more advanced implementations, only Ridge Regression and LightGBM found slight improvements across all metrics. These models were able to find patterns among monthly weather variables that assisted in describing how $PM_{2.5}$ was mapped out over 2022. The only other positive change from the added dataset is seen in the MAE value of the XGBoost model, which improved from 1.226 to 1.216, however it came at the expense of the other metrics. In short, ERA5's usefulness was dependent on the model to which it was applied. For some approaches it offered descriptive metrics, for others it simply provided noise.

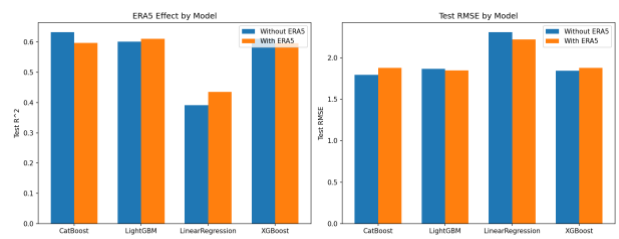

Figure 9 – Model ERA Comparison (R^2 and RMSE)

The bar chart above compares the differences between model performance with and without ERA support. The R^2 and RMSE values improved for both LightGBM and Linear Regression but worsened in XGBoost and CatBoost. This indicates that while there are beneficial predictors within the ERA data, further feature engineering would likely be required to increase its compatibility with the other gradient boosting methods. Further, LightGBM's improvement is largely in part to its ability to handle big data, making its forecasting output even more trustworthy.

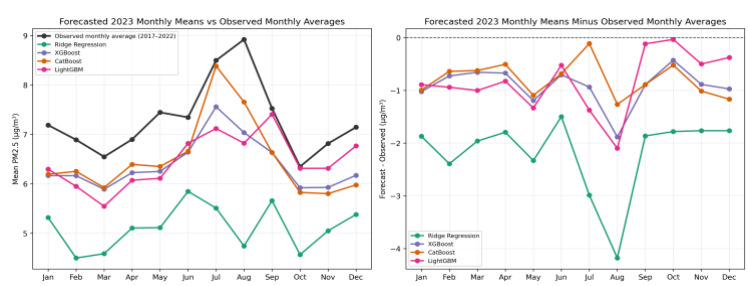

Figure 10 – 2023 Forecasted Monthly $PM_{2.5}$ Concentration Averages Compared to Historical Levels (2017-2022)

In Figure 10 above, our forecasted 2023 $PM_{2.5}$ monthly averaged levels for each model can be seen, as they compare to the historical averaged monthly levels between 2017 and 2022. In the graph, our gradient boosting models follow relative historical patterns but are noticeably more conservative in predicted concentrations. This can be attributed to a plethora of explanations, such as $PM_{2.5}$ changing significantly yearly from events like wildfires[1], emission changes[8], and others like COVID-19[6]. Regardless, it is an important affirmation that our gradient boosting models were able to map out very similar seasonality patterns of $PM_{2.5}$ pollution.

Additionally, looking at the graph on the right-side of Figure 10, it is clear that different models performed better through different parts of the year. CatBoost appeared to outperform the other models throughout the first 7 months of 2023, whereas LightGBM performed better in the latter part of the year. Building on that, XGBoost offered a more consistent performance throughout the year. However, Ridge Regression was clearly unable to compete with the other forecasting models, which is clear in it predicting lower concentrations over the summer when higher $PM_{2.5}$ is generally more common. Further, $PM_{2.5}$ increases within sunnier climates as the sun serves as a catalyst, speeding up the chemical reactions that can form these particles[9]. Therefore, it is important that our models would mimic this well-known pattern, showing larger levels during the summer and lower in the winter.

b. Conclusions

In conclusion, our study aimed to address the current gap in forecasting $PM_{2.5}$ pollution both in terms of overall scale, as well as generalizability. The inclusion of meteorological information was also questioned as to whether it helped or distracted from accurately forecasting $PM_{2.5}$ throughout North America. Through the findings discussed in the section prior, it was found that CatBoost was the best performing model when trained only on $PM_{2.5}$ measurements (No ERA5). Once the ERA data was added, LightGBM was found to outperform the other models. Out of the four models, only LightGBM and Ridge Regression were able to realize any improvement in the meteorological predictors.

Overall, while there is potential for the weather data to offer significant improvement, the best performing model for making historical predictions was CatBoost when run solely on the $PM_{2.5}$ dataset. Its final outputs are as follows, RMSE: 1.793, MAE: 1.159, and R^2: 0.633, which were the best across both scenarios. In terms of determining the best forecasting model, however, it would require direct $PM_{2.5}$ monthly measurements across North America for 2023 to make comparisons. While clear conclusions can be made about the model that best performed on the 2022 testing data, there was no clear model outperforming the rest among the forecasts. However, it can be concluded that based on historical averaged concentrations, CatBoost tended to deliver the best results in the first half of the year, while LightGBM excelled over the last half. It is also substantial that CatBoost was best able to capture the historical peak in $PM_{2.5}$ concentrations over the summertime. Meanwhile, XGBoost provided a consistent level of output throughout the entire period.

In a broader sense, our research enables individuals to understand the ways in which different gradient boosting methods can predict $PM_{2.5}$ concentrations throughout North America, both historically and through forecasting. The findings discussed in this paper can assist in benefitting policy, future monitoring, or preventative action[10].

VII. FUTURE WORK

While able to achieve much of what our research set out to accomplish there is still room for improvement:

The first main improvement that could be considered would be to include daily $PM_{2.5}$ measurements and ERA5 data[5], rather than our monthly values. This inclusion would allow our models to recognize more temporally detailed patterns that may allow for more accurate predictions. Some cons of this approach, however, are that the dataset would be around 40 gigabytes per year, and the training times would be exponentially greater. The computational cost would likely be so great that more advanced hardware would be required to make worthwhile daily $PM_{2.5}$ predictions. It was hypothesized that while accuracy would likely increase, the added time component would drastically increase costs.

The second potential improvement would be to produce more focused and interpretable predictions. This could mean forecasting $PM_{2.5}$ concentrations for cities or small regions and doing so in a way that is easily interpretable by the general public. A way in which this could be accomplished would be to implement Gradient Boosting Regression, which essentially acts as a simpler XGBoost model in terms of both methodology and explainability. This model does particularly well when focused on smaller datasets, opening the door for regional studies and increased understanding about how specific areas can monitor their emissions. It was hypothesized that this approach could help inspire directed efforts to monitoring, preparing, and even preventing $PM_{2.5}$ pollution. While its findings would be easier to understand, there is a significant loss in handling complex variables, and the model would only succeed depending on how large the chosen region was.

The third potential approach could be to develop a meta model[11] that essentially stacks the best models on top of each other to boost overall performance. This is done by first repeating the creation pipeline of training multiple models and creating historical predictions. Next, those prediction outputs are reused as the inputs for an outer layer model. That

top layer model then learns when and where different models perform best, this allows for overall accuracy to be optimized and increased. Some cons with this approach would again be computational cost, as well as an increased risk of overfitting. If the meta model only favors accuracy and uses the best model in every scenario, it will be unable to generalize its predictions to unknown values, especially forecasting frameworks. It was hypothesized that predictive accuracy has a high potential to increase, however, the risks associated with computational complexity and overfitting would increase as well.

Another potential improvement to this research could be the addition of forecasted weather information. While attempts were made to include weather data in terms of historical predictions, it was determined that the reanalysis data was not a realistic fit for forecasting frameworks. Therefore, the inclusion of this forecasted data would enable the models to access and understand how factors like wind, temperature, cloud cover, etc., impact $PM_{2.5}$ concentrations. It was hypothesized that if this data was incorporated cleanly it would help the models to understand more about the future levels of $PM_{2.5}$, in turn boosting their overall accuracy.

VIII. CODE AVAILABILITY

All the code is easily accessible and located on our public GitHub https://github.com/aliljas/DS340_Alfred_Otis

IX. DATA AVAILABILITY

Our overarching Monthly $PM_{2.5}$ dataset was taken from our parent paper[1], as listed in their data availability section [https://zenodo.org/records/10800980]. The second dataset used, titled ERA5[5], is from the ECMWF reanalysis, containing a diverse collection of weather values. To download this data, you must make an account on this website: https://cds.climate.copernicus.eu/. Then click on "ERA5 monthly averaged data on single levels from 1940 to present": [ERA5 Monthly Averaged Dataset]. Next, you download and subsetted the data to our chosen North America boundary of North: 84°, West: -168°, South: 7°, East: -52°. You then include variables of 10m u and 10m v-components of wind, 2m dewpoint temperature, 2m temperature, Surface pressure, Total precipitation, and Total cloud cover. Last, you set your date range from 2017-2022 at time 0:00, clicking every month, and storing your data as an unarchived NetCDF4 file. The final datasets are then placed in the raw data folder to be processed.

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